

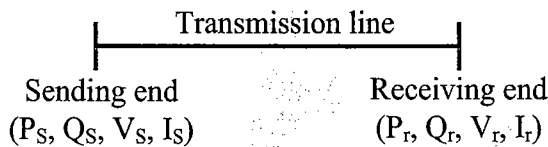


2.3 Steady State Analysis of Transmission Lines

There are two performance indices for any power apparatus

1. Efficiency —
 - Power Efficiency
 - Energy Efficiency
2. Regulation —
 - Voltage Regulation
 - Speed Regulation

The transmission line is a static device. So the performance of the transmission line is analyzed by considering the power efficiency and voltage regulation.



Power Efficiency:

$$\text{Efficiency} = \frac{\text{out put}}{\text{out put} + \text{losses}} \times 100\%$$

$$\eta = \frac{P_r}{P_r + 3I^2 R} \times 100\%$$

The losses in transmission line generally very less. So the efficiency will be very high i.e. (95-99%). So there is no much concern about the efficiency.

Voltage Regulation: It is defined as the change in magnitude of receiving end voltage from no load to full load while keeping the sending and voltage and the frequency of supply remains same.

$$\varepsilon = \frac{|V_{ro}| - |V_r|}{|V_r|} \times 100$$

V_{ro} = no load receiving voltage
 V_r = full load receiving voltage

Regulation means voltage drop. Percentage regulation is having good meaning when compared to regulation. Regulation depends on the quantity of load placed at receiving end and power factor of the load.

Representation of transmission lines: The transmission can be represented based on the length in which the power is carried out.

A) short transmission lines:- less than 80 km.

B) medium transmission line: 80 km to 160 km.

C) long transmission line: Above 160 km.

Short transmission line:

Generally charging current is very less in short transmission line so capacitance can be removed from the network. So it will be represented as RL lumped parameter model.

Medium Transmission lines: As the length is medium, charging is appreciable so capacitance will be considered in the network. So it will be represented as RLC lumped model.

Load end Capacitance model: The total shunt capacitance is placed at the receiving end only. (or) the total charging current will be allowed till the end of transmission line.

Sending and Capacitance model: The total shunt capacitance is at the sending end. There is no charging current in the transmission line.

Nominal – T model: The shunt capacitance is placed at the middle (or) the total charging current will be allowed upto middle of the transmission line.

Nominal – π model: The shunt capacitance is placed on both sides with equal magnitudes (or) the charging current of 50 percent magnitude is allowed till the end of transmission line.



Long Transmission line: It will be represented in RLCG distributed parameter model.

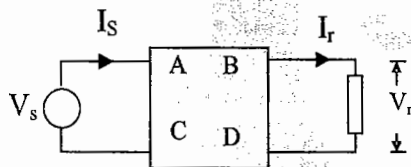
Uniform Distributed Parameters-These parameters are physical and electrically not separable.

Uniform distributed parameters are considered to evaluate the transient behavior of long transmission lines. i.e. switch closed condition. Exact mathematical solution is considered to evaluate the sending end voltage and current. Lumped parameters are considered to evaluate the steady state behavior of long transmission lines. The two possible network configurations are.

- (i) Equivalent – T
- (ii) Equivalent – π

The most and effective way of representing the transmission line is using two port network configuration port means pair of terminals.

ABCD parameters or Generalized circuit constants:



V_s, I_s are dependent values
 V_r, I_r are independent values

The dependent values are expressed in terms of independent values. With certain parameters and those parameters are called Transmission parameters or ABCD parameters.

$$V_s = AV_r + BI_r \dots (1)$$

$$I_s = CV_r + DI_r \dots (2)$$

$$\begin{pmatrix} V_s \\ I_s \end{pmatrix} = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \begin{pmatrix} V_r \\ I_r \end{pmatrix}$$

Equation (1) and (2) are standard transmission line equations. Parameters will be calculated by the following procedure.

Case-1:

Open circuit at the receiving end ($I_r = 0$)

$$V_s = A.V_{r0}, \quad I_s = C.V_{r0}$$

$$A = \frac{V_s}{V_{r0}}, \text{ Reverse voltage gain under OC (unitless quantity)}$$

$$C = \frac{I_s}{V_{r0}}, \text{ transfer admittance under OC (mho)}$$

$$\frac{A}{C} = \frac{V_s}{I_s} \bigg|_{I_r=0}, \text{ OC input impedance (Z}_{OC})$$

Case 2:

Short circuit at the receiving end ($V_r = 0$)

$$V_s = BI_{rsc}, \quad I_s = D.I_{rsc}$$

$$B = \frac{V_s}{I_{rsc}}, \text{ transfer impedance under SC } (\Omega)$$

$$D = \frac{I_s}{I_{rsc}}, \text{ Reverse current gain under SC (unitless quantity)}$$

$$\frac{B}{D} = \frac{V_s}{I_s} \bigg|_{V_r=0}, \text{ SC input impedance (Z}_{SC})$$

Characteristic impedance of transmission line

$$Z_C = \sqrt{Z_{oc} Z_{sc}}, \quad Z_C = \sqrt{\frac{AB}{CD}}$$

For a symmetrical transmission line $A = D$

For a reciprocal transmission line $AD - BC = 1$

$$\text{For symmetrical line } Z_C = \sqrt{\frac{B}{C}}$$

Example 2.3.1:

If the power transmission line is modeled so as to relate the sending end voltage V_s and current I_s to the receiving end voltage V_r and current I_r by the relationship

$$V_s = AV_r + BI_r, \quad I_s = CV_r + DI_r$$

Then which one of the following expresses the output impedance of the transmission line? (IAS-03)

- (a) $(DV_s - BI_s)/(-CV_s + AI_s)$
- (b) $(CV_s + AI_s)/(DV_s + BI_s)$
- (c) $(DV_s - CI_s)/(DV_s + BI_s)$
- (d) $(DV_s + BI_s)/(AV_s + DI_s)$

Sol: The given equations can be expressed as

$$\begin{pmatrix} V_s \\ I_s \end{pmatrix} = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \begin{pmatrix} V_r \\ I_r \end{pmatrix}$$

$$\begin{pmatrix} V_r \\ I_r \end{pmatrix} = \begin{pmatrix} A & B \\ C & D \end{pmatrix}^{-1} \begin{pmatrix} V_s \\ I_s \end{pmatrix}$$

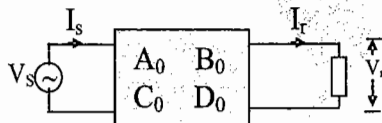
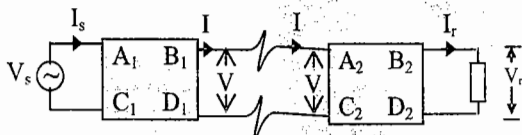
$$= \frac{1}{AD - BC} \begin{bmatrix} D & -B \\ -C & A \end{bmatrix} \begin{bmatrix} V_s \\ I_s \end{bmatrix}$$

$$= \begin{bmatrix} D & -B \\ -C & A \end{bmatrix} \begin{bmatrix} V_s \\ I_s \end{bmatrix}$$

The output impedance of transmission line is

$$\frac{V_r}{I_r} = \frac{DV_s - BI_s}{-CV_s + AI_s}$$

Transmission lines in series (or) Cascade (or) Tandem:

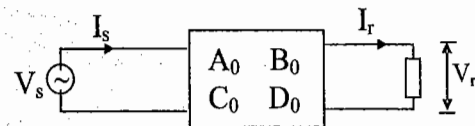
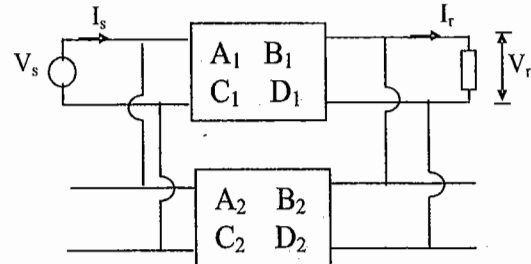


$$\begin{pmatrix} V_s \\ I_s \end{pmatrix} = \begin{pmatrix} A_0 & B_0 \\ C_0 & D_0 \end{pmatrix} \begin{pmatrix} V_r \\ I_r \end{pmatrix}$$

$$\begin{bmatrix} A_0 & B_0 \\ C_0 & D_0 \end{bmatrix} = \begin{bmatrix} A_1 & B_1 \\ C_1 & D_1 \end{bmatrix} \begin{bmatrix} A_2 & B_2 \\ C_2 & D_2 \end{bmatrix}$$

The resultant ABCD parameters of cascade combination of two networks is product of two ABCD vectors.

Two Transmission lines in parallel:-



$$A_0 = \frac{A_1 B_2 + A_2 B_1}{B_1 + B_2}$$

$$B_0 = \frac{B_1 B_2}{B_1 + B_2}$$

$$C_0 = (C_1 + C_2) + \frac{(A_1 - A_2)(D_2 - D_1)}{B_1 + B_2}$$

$$D_0 = \frac{D_1 B_2 + D_2 B_1}{B_1 + B_2}$$

If two lines are identical transmission lines

$$A_0 = A, \quad B_0 = B/2, \quad C_0 = 2C \quad \text{and} \quad D_0 = D$$

Example 2.3.2:

Two identical lines having $A = 0.93 \angle 1.2^\circ$, $B = 132 \angle 72^\circ$, and

$C = 10^{-3} \angle 90^\circ$ and $D = 0.93 \angle 1.2^\circ$ are connected in parallel. The equivalent transmission line parameters are

(a) $A = D = 0.93 \angle 1.2^\circ$, $B = 264 \angle 72^\circ$, and $C = 5 \times 10^{-4} \angle 90^\circ$

(b) $A = D = 1.86 \angle 1.2^\circ$, $B = 132 \angle 72^\circ$, and $C = 2 \times 10^{-3} \angle 90^\circ$

(c) $A = D = 0.93 \angle 1.2^\circ$, $B = 66 \angle 72^\circ$, and $C = 2 \times 10^{-3} \angle 90^\circ$

(d) $A = D = 0.93 \angle 1.2^\circ$, $B = 132 \angle 72^\circ$, and $C = 10^{-3} \angle 90^\circ$



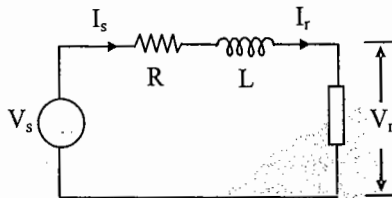
Sol: The resultant ABCD of this parallel combination

$$A = D = 0.93 \angle 1.2^\circ$$

$$B = 66 \angle 72^\circ \text{ ohm}$$

$$C = 2 \times 10^{-3} \angle 90^\circ \text{ mho}$$

Short transmission line: Series combination of resistance and Inductance.



$$Z = R + jX$$

$$V_s = 1.0 V_r + Z I_r$$

$$I_s = 0. V_r + 1.0 I_r$$

$$\begin{pmatrix} A & B \\ C & D \end{pmatrix} = \begin{pmatrix} 1 & Z \\ 0 & 1 \end{pmatrix}$$

$A = D$, network is symmetrical

$AD - BC = 1$, network is reciprocal

Regulation:

$$\% \epsilon = \frac{|V_{ro}| - |V_r|}{|V_r|} \times 100\%$$

on no-load $I_r = 0$ so $V_s = AV_{ro}$

$$|V_{ro}| = \frac{|V_s|}{A}$$

$$\% \epsilon = \frac{\left| \frac{V_s}{A} \right| - |V_r|}{|V_r|}$$

In case of short transmission line $|A| = 1.0$

$$|V_{ro}| = |V_s|$$

$$\% \epsilon = \frac{|V_s| - |V_r|}{|V_r|} \times 100$$

The sending end voltage magnitude of short transmission line will be represented as

$$|V_s| = |V_r| + I_r R \cos \phi_r \pm I_r X \sin \phi_r$$

$$\% \epsilon = \frac{|V_s| - |V_r|}{|V_r|} \times 100$$

$$\% \epsilon = \frac{I_r R \cos \phi_r \pm I_r X \sin \phi_r}{V_r} \times 100\%$$

'+' stands for lagging p.f

'-' stands for leading p.f

$$\% \epsilon = \% R \cos \phi_r \pm \% X \sin \phi_r$$

For lagging power factor always voltage regulation is positive

For leading power factor the regulation will be either positive or negative or zero.

The regulation will be maximum (positive) at

$$\phi_r = \tan^{-1} \left(\frac{X}{R} \right)$$

$\phi_r = \theta$ (impedance angle of transmission line)

- The regulation will be zero at leading power factor,

$$\phi_r = \tan^{-1} \left(\frac{R}{X} \right)$$

$$\phi_r = \frac{\pi}{2} - \theta$$

Example 2.3.3:

A distribution line having series impedance in per unit $0.08 + j0.05$. What is the voltage regulation with a load power factor of 0.8 lagging at half pull load condition

Sol: at FL, %Voltage regulation

$$= \% R \cos \phi_r + \% X \sin \phi_r$$

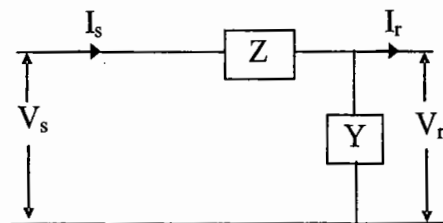
$$= (8 \times 0.8 + 5 \times 0.6)\%$$

$$= 9.4 \%$$

$$\text{At } \frac{1}{2} \text{ FL, \%Voltage regulation} = \frac{1}{2} \times 9.4\% = 4.7 \%$$

Medium Transmission line

Receiving end capacitor model:





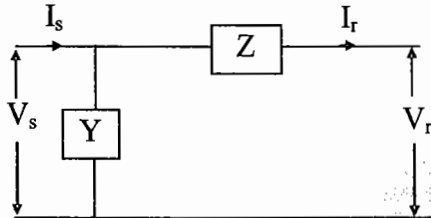
$$\begin{pmatrix} V_s \\ I_s \end{pmatrix} = \begin{pmatrix} 1 & Z \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ Y/2 & 1 \end{pmatrix} \begin{pmatrix} V_r \\ I_r \end{pmatrix}$$

$A = 1 + ZY$, $B = Z$, $C = Y$ and $D = 1.0$

$A \neq D$, network is unsymmetrical

$AD - BC = 1$ network is reciprocal

Sending end capacitor model



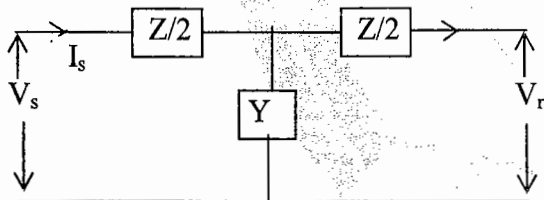
$$\begin{pmatrix} V_s \\ I_s \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ Y & 1 \end{pmatrix} \begin{pmatrix} 1 & Z \\ 0 & 1 \end{pmatrix} \begin{pmatrix} V_r \\ I_r \end{pmatrix}$$

$A = 1.0$, $B = Z$, $C = Y$ and $D = 1 + ZY$

$A \neq D$, network is unsymmetrical

$AD - BC = 1$ network is reciprocal

Nominal - T Network:-



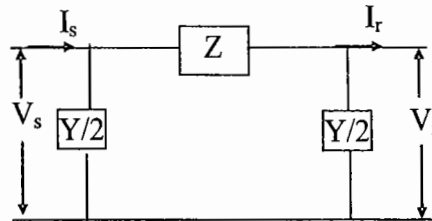
$$\begin{pmatrix} V_s \\ I_s \end{pmatrix} = \begin{pmatrix} 1 & Z/2 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ Y & 1 \end{pmatrix} \begin{pmatrix} 1 & Z/2 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} V_r \\ I_r \end{pmatrix}$$

$$\begin{pmatrix} V_s \\ I_s \end{pmatrix} = \begin{pmatrix} 1 + \frac{ZY}{2} & Z \left(1 + \frac{ZY}{4}\right) \\ Y & 1 + \frac{ZY}{2} \end{pmatrix} \begin{pmatrix} V_r \\ I_r \end{pmatrix}$$

$A = D$, network is symmetrical

$AD - BC = 1$, network is reciprocal

Nominal - π model:



$$\begin{pmatrix} V_s \\ I_s \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ Y/2 & 1 \end{pmatrix} \begin{pmatrix} 1 & Z \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ Y/2 & 1 \end{pmatrix} \begin{pmatrix} V_r \\ I_r \end{pmatrix}$$

$$\begin{pmatrix} V_s \\ I_s \end{pmatrix} = \begin{pmatrix} 1 + \frac{ZY}{2} & Z \\ Y \left(1 + \frac{ZY}{4}\right) & 1 + \frac{ZY}{2} \end{pmatrix} \begin{pmatrix} V_r \\ I_r \end{pmatrix}$$

$A = D$, network is symmetrical

$AD - BC = 1$, network is reciprocal

For all the networks the preferable voltage

regulation formula is $\% \varepsilon = \frac{\left| \frac{V_s}{A} \right| - |V_r|}{|V_r|}$

The Nominal T Network is a symmetric and reciprocal network.

Sending end capacitance method is an uncompensated method. Receiving end capacitor method is an over compensated method. So the medium transmission lines are represented either by nominal - π (or) nominal - T methods. However the nominal - π is preferred on according to load conditions.

Example 2.3.4:

A medium line with parameters A, B, C, D is extended by connecting a short line of impedance Z in series. The overall ABCD parameters of the series combination will be (IAS-03)

- $A, AZ, C + D/Z, D$
- $A, AZ + B, C, CZ + D$
- $A + BZ, B, C + DZ, D$
- $AZ, B, C/Z, D$



Sol: Medium line and short line are cascaded together the resultant ABCD parameters as follows

$$\begin{bmatrix} A_0 & B_0 \\ C_0 & D_0 \end{bmatrix} = \begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} 1 & Z \\ 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} A & AZ + B \\ C & CZ + D \end{bmatrix}$$

Example 2.3.5:

A 50Hz 3-phase transmission line of length 100 km has a capacitance of $0.03/\pi \mu\text{F}$ per km. It is represented as a π model. The shunt admittance at each end of the transmission line will be

(a) $150 \times 10^{-6} \angle 90^\circ \text{ mho}$

(b) $100 \times 10^{-6} \angle 90^\circ \text{ mho}$

(c) $50 \times 10^{-6} \angle 90^\circ \text{ mho}$

(d) $\frac{10^6}{150} \angle 90^\circ \text{ mho}$

Sol: Total shunt admittance,

$$Y = j\omega C/\text{km} \cdot l$$

$$= j 2 \pi \times 50 \times \frac{0.03}{\pi} \times 100 \mu\text{S}$$

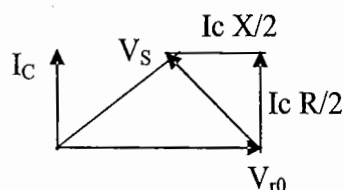
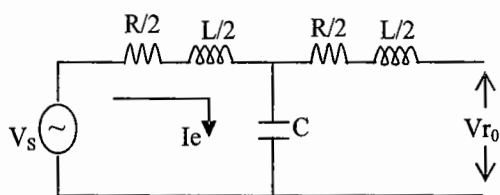
$$= 300 \angle 90^\circ \mu\text{S}$$

In the nominal π model half of the admittance placed at sending end side and remaining half placed at receiving end side.

The shunt admittance at each end of

$$\text{transmission line is } \frac{Y}{2} = 150 \times 10^{-6} \angle 90^\circ \text{ mho}$$

Ferranti Effect: When the transmission line operating at no-load (or) light load condition, the receiving end voltage is more than the sending voltage. This phenomena is called ferranti effect. It is more severe in long transmission line.



Example 2.3.6:

The ABCD constants of a 3 phase transmission line are $A = D = 0.8 \angle 1^\circ$, $B = 170 \angle 80^\circ \Omega$, $C = 0.002 \angle 90.4^\circ \text{ mho}$. The sending end voltage is 400KV. The receiving end voltage under no load condition is

(APGENCO)

(a) 400 kV

(b) 500 kV

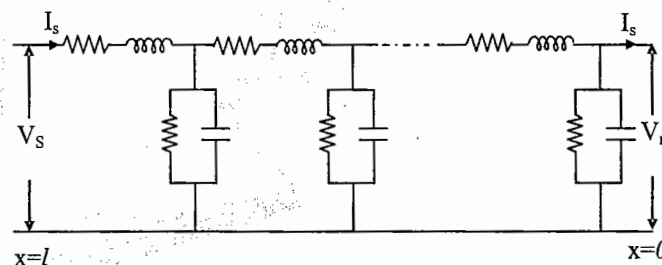
(c) 320 kV

(d) 417 kV

Sol: No load receiving end voltage

$$\begin{aligned} |V_{r0}| &= \left| \frac{V_s}{A} \right| \\ &= \frac{400}{0.8} \text{ kV} \\ &= 500 \text{ kV} \end{aligned}$$

Long Transmission line or Distributed model ABCD parameters:



Series impedance per km, $z = r + j\omega L \Omega/\text{km}$

Shunt admittance per km, $y = g + j\omega C \text{ mho}/\text{km}$

At any point x on the line voltage and currents will be given as

$$V_x = (\cosh \gamma x) V_r + (Z_c \sinh \gamma x) I_r$$

$$I_x = \left(\frac{1}{Z_c} \sinh \gamma x \right) V_r + (\cosh \gamma x) I_r$$

' γ ' called as propagation constant, $\gamma = \sqrt{zy}$

' Z_c ' called as characteristic impedance, $Z_c = \sqrt{\frac{z}{y}}$



At the sending end side ($x = l$), $V_x = V_s$ and $I_x = I_s$
So the sending end voltage and currents given as

$$V_s = (\cosh \gamma l) V_r + (Z_c \sinh \gamma l) I_r$$

$$I_s = \left(\frac{1}{Z_c} \sinh \gamma l \right) V_r + (\cosh \gamma l) I_r$$

$$\begin{bmatrix} V_s \\ I_s \end{bmatrix} = \begin{bmatrix} \cosh \gamma l & Z_c \sinh \gamma l \\ \frac{1}{Z_c} \sinh \gamma l & \cosh \gamma l \end{bmatrix} \begin{bmatrix} V_r \\ I_r \end{bmatrix}$$

These are the exact ABCD parameters (or) ABCD parameters of distributed parameter model (or) (or) practical transmission line model.

Here $A = D \Rightarrow$ Distributed mode is symmetrical

$AD - BC = 1 \Rightarrow$ network is reciprocal

Approximate ABCD parameters of long line:

$$\cosh \gamma l = 1 + \frac{(\gamma l)^2}{2!} + \frac{(\gamma l)^4}{4!} + \dots$$

$$\sinh \gamma l = \gamma l + \frac{(\gamma l)^3}{3!} + \frac{(\gamma l)^5}{5!} + \dots$$

$$\cosh \gamma l \cong 1 + \frac{(\gamma l)^2}{2!}$$

$$\sinh \gamma l \cong \gamma l + \frac{(\gamma l)^3}{3!}$$

$$(\gamma l)^2 = (\sqrt{zy} l)^2$$

$$= ZY, (Z = zl, Y = yl)$$

$$A = D = \cosh \gamma l$$

$$= 1 + \frac{(\gamma l)^2}{2}$$

$$= 1 + \frac{ZY}{2}$$

$$B = Z_c \sinh \gamma l$$

$$= Z_c \gamma l \left[1 + \frac{(\gamma l)^2}{6} \right]$$

$$= \sqrt{\frac{Z}{y}} \sqrt{zy} l \left[1 + \frac{zy}{6} \right]$$

$$B = Z \left[1 + \frac{ZY}{6} \right]$$

$$C = \frac{1}{Z_c} \sinh \gamma l = \frac{1}{Z_c} \gamma l \left[1 + \frac{(\gamma l)^2}{6} \right]$$

$$= Y \left[1 + \frac{ZY}{6} \right]$$

Conclusion on ABCD parameters:

ABCD parameters will be represented as

$$A = |A| \angle \alpha,$$

$$B = |B| \angle \beta,$$

$$C = |C| \angle \gamma,$$

$$D = |D| \angle \Delta$$

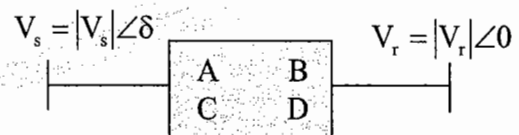
Generally

- $|A| \leq 1$, very close to 1, $\alpha = 0$ to 10°
- Angle β is equivalent to impedance angle of line such that $\beta = 60^\circ$ to 90° . $|B|$ value depends on length of the line
- Angle $\gamma \approx 90^\circ$. $|C|$ depends on length of the line
- $|D| \leq 1$, very close to 1, $\Delta = 0$ to 10°

If length of transmission line increases then

- $|A| \downarrow, \alpha \uparrow$
- $|B| \uparrow, \beta$ almost constant
- $|C| \uparrow, \gamma$ almost constant
- $|D| \downarrow, \Delta \downarrow$

Static Power Transfer Equations in terms of ABCD parameters



The angular difference between sending end and receiving end bus voltages is known as load angle of transmission line.

The receiving end power transfer

$$S_r = V_r I_r^*$$

(Always complex power taken as VI^* provided that inductive load absorbs reactive power and capacitive load delivers reactive power)



$$S_r = V_r \left(\frac{V_s - AV_r}{B} \right)^*$$

$$S_r = |V_r| \angle 0 \left(\frac{|V_s| \angle \delta - |A| \angle \alpha |V_r| \angle 0}{|B| \angle \beta} \right)^*$$

$$S_r = \frac{|V_s| |V_r|}{|B|} \angle (\beta - \delta) - \frac{|A|}{|B|} |V_r|^2 \angle (\beta - \alpha)$$

$$P_r = \frac{|V_s| |V_r|}{|B|} \cos(\beta - \delta) - \frac{|A|}{|B|} |V_r|^2 \cos(\beta - \alpha) \quad \text{---(1)}$$

$$Q_r = \frac{|V_s| |V_r|}{|B|} \sin(\beta - \delta) - \frac{|A|}{|B|} |V_r|^2 \sin(\beta - \alpha) \quad \text{---(2)}$$

The loci of the above equations are circle. For fixed receiving end voltages they will be concentric circles.

The above equations (1) & (2) are non linear algebraic equations.

Condition for maximum power transfer

Maximum power transfer occurs for $\frac{dP_r}{d\delta} = 0$

$$\text{Where } \frac{dP_r}{d\delta} = \frac{|V_s| |V_r|}{|B|} \sin(\beta - \delta)$$

So $P_{\max}$ occurs for $\delta = \beta$

$$P_{\max} = \frac{|V_s| |V_r|}{|B|} - \frac{|A|}{|B|} |V_r|^2 \cos(\beta - \alpha) \quad \text{---(3)}$$

Q_r value during $P_{\max}$ will be given as,

$$Q_r = -\frac{|A|}{|B|} |V_r|^2 \sin(\beta - \alpha) \quad \text{--- (4)}$$

It indicates that the load at the receiving end is capacitive which impractical.

Generally for transient stability the operating angles of the system to deliver the required real power is 30 - 45°.

For Short transmission line:

$$|A| = 1.0, \alpha = 0, |B| = |Z| \text{ and } |B| = \theta$$

Real and Reactive power flow expressions are

$$P_r = \frac{|V_s| |V_r|}{|Z|} \cos(\theta - \delta) - \frac{|V_r|^2}{|Z|} \cos(\theta)$$

$$Q_r = \frac{|V_s| |V_r|}{|Z|} \sin(\theta - \delta) - \frac{|V_r|^2}{|Z|} \sin(\theta)$$

$P_{\max}$ occurs for $\delta = \theta$

$$P_{\max} = \frac{|V_s| |V_r|}{|Z|} - \frac{|V_r|^2}{|Z|} \cos(\theta)$$

Reactive power during $P_{\max}$,

$$Q_r = -\frac{|V_r|^2}{|Z|} \sin \theta$$

Approximate Power transfer equations

- For long transmission line $R \ll X$, so R can be neglected
- Shunt admittance of transmission line will be neglected

With these assumptions,

$$|A| = 1.0, \alpha = 0, |Z| = X \text{ and } \theta = 90^\circ$$

So the real and reactive power equations are modified as

$$P_r = \frac{|V_s| |V_r|}{X} \sin \delta$$

$$P_r = P_{\max} \sin \delta \quad \text{.... (5)}$$

$$Q_r = \frac{|V_r|}{X} [|V_s| \cos \delta - |V_r|]$$

$P_{\max}$ occurs for $\delta = 90^\circ$

$$P_{\max} = \frac{|V_s| |V_r|}{X}$$

Q_r during $P_{\max}$

$$Q_r = -\frac{|V_r|^2}{X} \text{ (negative)}$$

The real power transfer depends on power angle and the reactance power depends on the difference of voltage between the sending end and receiving end.



Example 2.3.7 :

A feeder with reactance of 0.2 p.u has a sending-end voltage of 1.2 p.u. If the reactive power supplied at the receiving end of the feeder is 0.3 p.u., the approximate voltage drop in the feeder will be _____

Sol:

$$X = 0.2 \text{ p.u}$$

$$|V_s| = 1.2 \text{ p.u}$$

$$Q_r = 0.3 \text{ p.u}$$

$$Q_r = \frac{|V_r|}{X} (|V_s| - |V_r|)$$

$$0.3 = \frac{|V_r|}{0.2} (1.2 - |V_r|)$$

$$0.06 = 1.2 |V_r| - |V_r|^2$$

$$|V_r|^2 - 1.2 |V_r| + 0.06 = 0$$

$$|V_r| = 1.147 \text{ p.u}$$

$$\begin{aligned} \text{Voltage drop} &= |V_s| - |V_r| \\ &= 1.2 - 1.147 = 0.053 \text{ p.u} \end{aligned}$$

Sending end power transfer equation:

Complex power flow at sending end side given as

$$S_s = V_s I_s^*$$

$$S_s = \frac{D}{B} |V_s|^2 \angle(\beta - \Delta) - \frac{|V_s| |V_r|}{|B|} \angle(\beta + \delta)$$

The real and reactive powers at sending end are

$$P_s = \frac{D}{B} |V_s|^2 \cos(\beta - \Delta) - \frac{|V_s| |V_r|}{|B|} \cos(\beta + \delta)$$

$$Q_s = \frac{D}{B} |V_s|^2 \sin(\beta - \Delta) - \frac{|V_s| |V_r|}{|B|} \sin(\beta + \delta)$$

2.4 Transient Analysis on Long Transmission line

Long transmission line:- Uniform distributed parameters are considered.

When the switch is closed, the current will be allowed to flow through the inductor without any time delay, because the value of inductance is very less which is not sufficient to oppose the flow of current. When the supply voltage is between 0-180°, the inductor will also charge from 0-180° and the energy stored will be $\frac{1}{2} L I^2$. The magnetic field is produced. When the switched voltage between 180-360° the inductor will also charged from 180 - 360°. However it will discharge the energy stored in the previous cycle i/e 0-180 to the capacitor so that the static field is produced. The energy stored in capacitor is $\frac{1}{2} C V^2$. There is 180° phase difference in charging of inductor and capacitance. If the transmission line is loss-less line the voltage across the capacitor is equal to the switch voltage. Otherwise it will be less than the switched voltage.

